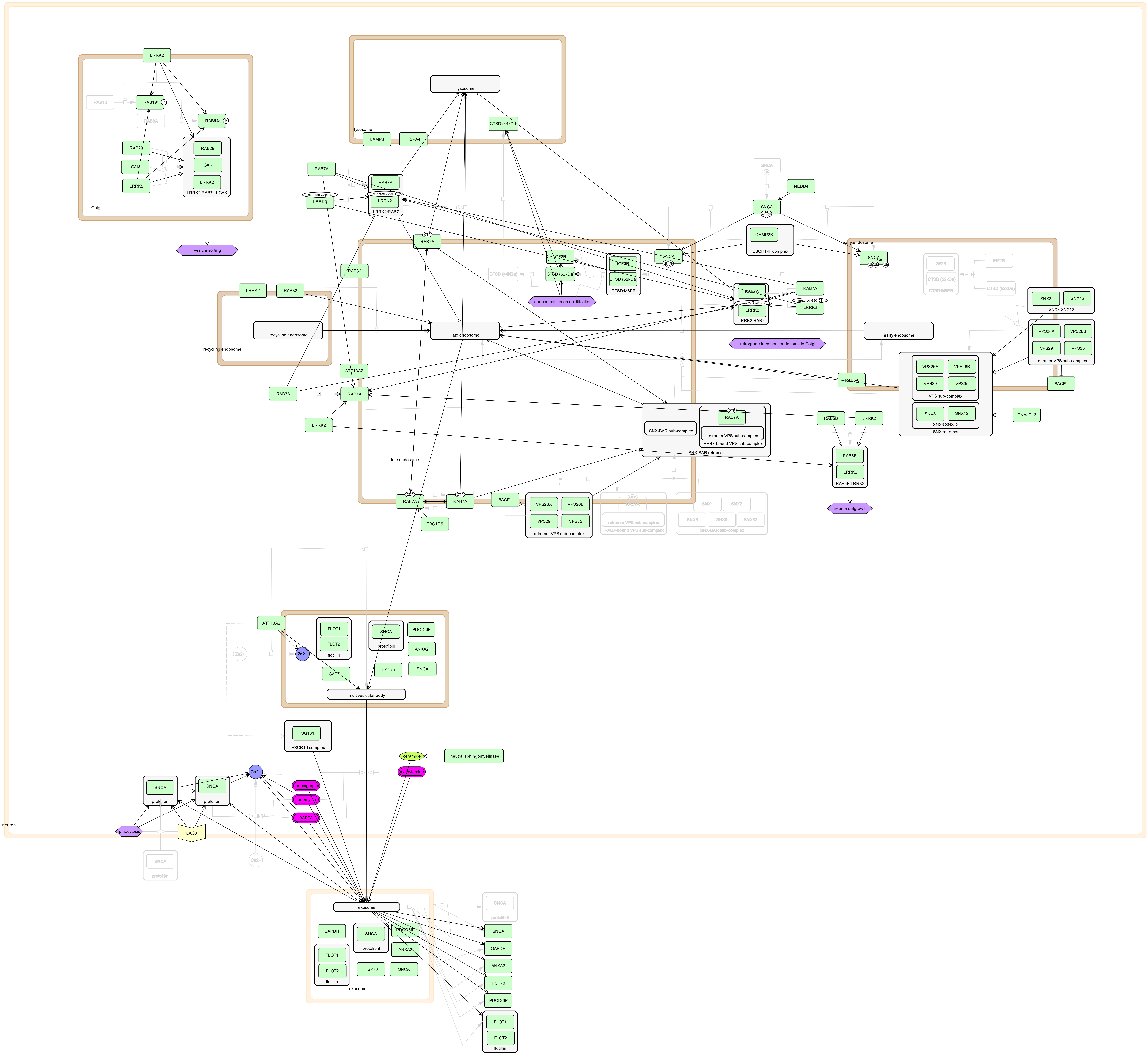
LRRK2
RAB10
RAB10
P
RAB8A
RAB8A
P
RAB29
GAK
LRRK2
RAB29
GAK
LRRK2
LRRK2:RAB7L1:GAK
Golgi
vesicle sorting
lysosome
lysosome
LAMP3
HSPA4
CTSD (44kDa)
RAB7A
mutated G2019S
LRRK2
RAB7A
mutated G2019S
LRRK2
LRRK2:RAB7
SNCA
Ub
NEDD4
SNCA
Ub
CHMP2B
ESCRT-III complex
early endosome
SNCA
S129
Ub
GTP
RAB7A
RAB32
IGF2R
CTSD (44kDa)
CTSD (52kDa)
IGF2R
CTSD (52kDa)
CTSD:M6PR
SNCA
Ub
endosomal lumen acidification
IGF2R
CTSD (52kDa)
CTSD:M6PR
IGF2R
CTSD (52kDa)
RAB7A
RAB7A
mutated G2019S
LRRK2
LRRK2:RAB7
mutated G2019S
LRRK2
LRRK2
RAB32
recycling endosome
recycling endosome
late endosome
early endosome
retrograde transport, endosome to Golgi
SNX3
SNX12
SNX3:SNX12
VPS26A
VPS26B
VPS29
VPS35
retromer VPS sub-complex
ATP13A2
RAB7A
RAB7A
?
LRRK2
RAB5A
VPS26A
VPS26B
VPS29
VPS35
VPS sub-complex
SNX3
SNX12
SNX3:SNX12
SNX retromer
BACE1
DNAJC13
SNX-BAR sub-complex
GTP
RAB7A
retromer VPS sub-complex
RAB7-bound VPS sub-complex
SNX-BAR retromer
RAB5B
LRRK2
RAB5B
LRRK2
RAB5B:LRRK2
neurite outgrowth
late endosome
GDP
RAB7A
GTP
RAB7A
TBC1D5
BACE1
VPS26A
VPS26B
VPS29
VPS35
retromer VPS sub-complex
GTP
RAB7A
retromer VPS sub-complex
RAB7-bound VPS sub-complex
SNX1
SNX2
SNX5
SNX6
SNX32
SNX-BAR sub-complex
ATP13A2
Zn2+
Zn2+
FLOT1
FLOT2
flotillin
SNCA
protofibril
PDCD6IP
ANXA2
GAPDH
HSP70
SNCA
multivesicular body
TSG101
ESCRT-I complex
ceramide
neutral sphingomyelinase
methylamine
Ca2+
thapsigargin
ionomycin
BAPTA
SNCA
protofibril
SNCA
protofibril
neuron
pinocytosis
LAG3
SNCA
protofibril
Ca2+
exosome
GAPDH
SNCA
protofibril
PDCD6IP
ANXA2
FLOT1
FLOT2
flotillin
HSP70
SNCA
exosome
SNCA
protofibril
SNCA
GAPDH
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HSP70
PDCD6IP
FLOT1
FLOT2
flotillin

GAK
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RAB29
GAK
LRRK2
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RAB8A
P
RAB10
P
RAB5B
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RAB7A
CHMP2B
ESCRT-III complex
NEDD4
SNCA
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neutral sphingomyelinase
methylamine
TSG101
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thapsigargin
SNCA
protofibril
BAPTA
ionomycin
ceramide
vesicle sorting
RAB5B
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RAB5B:LRRK2
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TBC1D5
RAB5A
early endosome
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VPS26B
VPS29
VPS35
retromer VPS sub-complex
SNX3
SNX12
SNX3:SNX12
SNCA
S129
Ub
DNAJC13
Ca2+
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neurite outgrowth
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LRRK2
LRRK2:RAB7
IGF2R
CTSD (52kDa)
late endosome
BACE1
SNX-BAR sub-complex
GTP
RAB7A
retromer VPS sub-complex
RAB7-bound VPS sub-complex
SNX-BAR retromer
GDP
RAB7A
LRRK2
recycling endosome
RAB32
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VPS26B
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VPS35
VPS sub-complex
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SNX12
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HSP70
SNCA
protofibril
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GAPDH
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HSP70
SNCA
FLOT1
FLOT2
flotillin
SNCA
protofibril
multivesicular body
FLOT1
FLOT2
flotillin
GAPDH
HSP70
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Zn2+